# Planck's Hypothesis and the Photoelectric Effect

PHYS 2203 — Introductory Physics III

October 19, 2026

## Overview

Last lecture, classical physics failed to explain the blackbody spectrum. Giving every standing wave in a hot cavity the same average energy $k_BT$ leads to the ultraviolet catastrophe. This lecture begins with Planck's solution of 1900, which assumes that the energy of each standing wave comes in discrete amounts $hf$. Planck regarded this as a property of the oscillating charges in the walls, not of light itself. We then turn to the photoelectric effect, in which light ejects electrons from a metal in ways that contradict the wave picture, and see how Einstein explained it in 1905 by proposing that light consists of particles, photons, each carrying energy $E = hf$.

## 1 Planck's Hypothesis

Lecture 21 ended with the failure of classical physics to explain the blackbody spectrum. Classically, the spectral energy density is the number of standing waves in each frequency range times the average energy of each wave, which the equipartition theorem sets to $k_BT$:

$$\frac{\mathrm{d}U}{\mathrm{d}f} = k_BT \times \frac{8\pi V}{c^3} f^2 \qquad \text{(classical prediction)}. \tag{1}$$

This agrees with experiment at low frequencies, but grows without limit at high frequencies: the ultraviolet catastrophe.

In 1900 Max Planck found a formula that matches the measured spectrum at every frequency and every temperature. To obtain it he made a radical assumption: the energy of a standing wave of frequency $f$ cannot take any value, but only whole-number multiples of a basic amount,

$$\boxed{E = n\,hf, \qquad n = 0, 1, 2, 3, \ldots} \tag{2}$$

where $h$ is a new constant of nature. A wave of frequency $f$ can gain or lose energy only in chunks of size $hf$.

The error in the classical theory is in the average energy per wave. Classically, the average is taken over a continuous range of possible energies, which gives $k_BT$. If only the discrete energies $nhf$ are allowed, the average comes out differently. Replacing $k_BT$ with Planck's result gives **Planck's spectral energy density**:

$$\boxed{\frac{\mathrm{d}U}{\mathrm{d}f} = \frac{hf}{e^{hf/k_BT} - 1} \times \frac{8\pi V}{c^3} f^2.} \tag{3}$$

This fits the measured curves perfectly.

The first factor, the average energy per wave, is what fixes the high-frequency problem. Compare the size of a single chunk, $hf$, with the thermal energy $k_BT$.

* At **low frequencies**, $hf \ll k_BT$. The chunks are tiny compared with the thermal energy, so the energy is effectively continuous. Using $e^x \approx 1 + x$ for small $x$,

$$\frac{hf}{e^{hf/k_BT} - 1} \approx \frac{hf}{hf/k_BT} = k_BT, \tag{4}$$

  and Planck's formula agrees with the classical one.

* At **high frequencies**, $hf \gg k_BT$. A single chunk is more than the thermal energy can usually supply, so these waves are rarely excited at all. The exponential in the denominator becomes huge, the average energy per wave falls toward zero much faster than $f^2$ grows, and the spectrum turns over and goes to zero, as observed.

The peak falls where $hf$ is comparable to $k_BT$. At a higher temperature larger chunks $hf$ can be excited, so the peak moves to higher frequency. This is Wien's law.

Planck's formula contains one new constant, $h$, and he determined its value by fitting the formula to the measured spectra. The formula then reproduces both Wien's law and Stefan's law, and predicts the values of their constants in terms of $h$ and $k_B$.

**Planck's constant.**

$$h = 6.63 \times 10^{-34}\ \text{J} \cdot \text{s} = 4.14 \times 10^{-15}\ \text{eV} \cdot \text{s}.$$

The appearance of $h$ in a result is the signature of quantum physics. Since photon energies are usually quoted in eV and wavelengths in nm, the combination

$$hc = 1240\ \text{eV} \cdot \text{nm}$$

is very convenient.

Planck regarded his assumption as a property of the oscillating charges in the walls, a mathematical device that made the formula work. He did not suggest that light itself comes in chunks. That step was taken by Einstein five years later, to explain the photoelectric effect. Planck received the 1918 Nobel Prize for the discovery of energy quanta.

## 2 The Photoelectric Effect

In 1887 Heinrich Hertz observed that light shining on a metal surface can eject electrons from it. This is the **photoelectric effect**. The electrons in a metal are bound to it, and a minimum energy, the **work function** $\phi$, is needed to free one. Any energy left over becomes the electron's kinetic energy. Some values are listed in Table 1.

| Metal | $\phi$ (eV) |
|---|---|
| Cesium | 1.9 |
| Potassium | 2.2 |
| Sodium | 2.3 |
| Magnesium | 3.7 |
| Zinc | 4.3 |
| Tungsten | 4.5 |

Table 1: Work functions of some metals.

**The experiment**

Figure 1 shows the apparatus. Light falls on a metal plate (the emitter) inside an evacuated tube. Ejected electrons travel to a second plate (the collector), and the ammeter measures the resulting current $i$. An adjustable voltage source applies a potential difference $V$ between the plates, measured by the voltmeter.

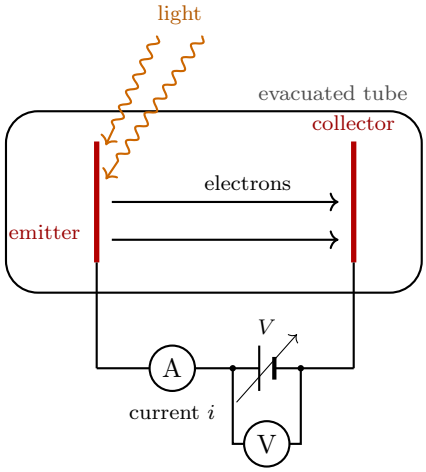



Figure 1: Photoelectric apparatus. Light ejects electrons from the emitter, and those reaching the collector produce a current $i$, measured by the ammeter (A). An adjustable source applies a potential difference $V$ between the plates, measured by the voltmeter (V).

If the collector is made *negative* relative to the emitter, the electrons are pushed back as they cross the gap. An electron (charge $-e$) moving through a potential difference $\Delta V$ changes its kinetic energy by $\Delta K = -(-e)\Delta V$, so crossing to a collector at potential $-V$ costs it kinetic energy $eV$. Only electrons that leave the emitter with $K > eV$ reach the collector. As $V$ is increased, the current drops, and it reaches zero at the **stopping voltage** $V_s$, when even the fastest electrons are turned back:

$$\boxed{K_{\max} = eV_s.} \tag{5}$$

Measuring $V_s$ therefore measures the maximum kinetic energy of the ejected electrons. In eV the two numbers are equal: a stopping voltage of 1.5 V means $K_{\max} = 1.5$ eV.

### What classical physics predicts

Classically, light is a wave whose energy is spread continuously over the wavefront, with intensity proportional to the square of the electric field amplitude ($I = \frac{1}{2}c\varepsilon_0 E_0^2$). An electron in the metal is shaken by the oscillating field and gradually absorbs energy until it can escape. This picture makes three predictions.

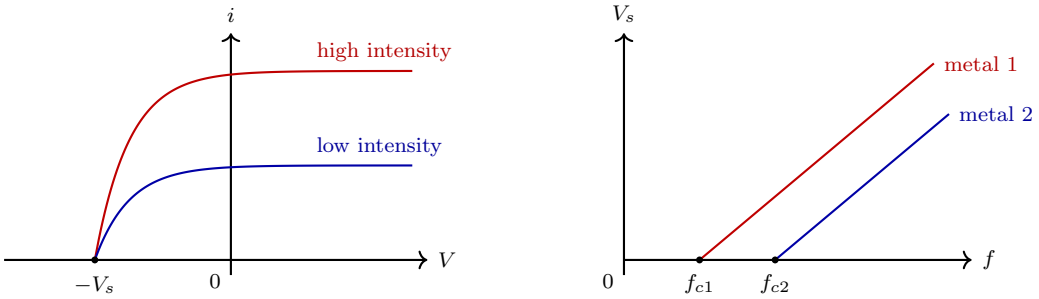
1. **Brighter light gives faster electrons.** A more intense wave has a stronger electric field, which should give the electrons more energy. $K_{\max}$ should increase with intensity.

2. **Any frequency should work.** Intensity does not depend on frequency, so light of any frequency, if intense enough, should eventually free electrons.

3. **Dim light should take time.** With very weak light, an electron must absorb energy for some time before it has enough to escape, so there should be a measurable delay between turning on the light and the first electrons.

### What is observed

The measurements are summarized in Figure 2.




(a) same stopping voltage for both intensities

(b) straight lines of equal slope, independent of intensity

Figure 2: (a) Photocurrent versus the potential $V$ of the collector relative to the emitter, for two intensities of the same light. The current is proportional to intensity, but it stops at the same stopping voltage. (b) Stopping voltage versus frequency for two metals. Each is a straight line starting at a cutoff frequency $f_c$, and the lines have the same slope.

Each classical prediction fails:

| Classical prediction | Observation |
|---|---|
| $K_{\max}$ increases with intensity. | $K_{\max}$ is independent of intensity. Brighter light ejects *more* electrons, not faster ones. |
| Any frequency works if the light is intense enough. | Below a **cutoff frequency** $f_c$, no electrons are ejected, no matter how intense the light. |
| Dim light should show a time delay. | Electrons appear immediately, within $10^{-9}$ s, even for very dim light. |

In addition, $K_{\max}$ increases *linearly* with frequency, with the same slope for every metal. Classical physics gives no reason for any of this.

## 3 Einstein's Photon Explanation

In 1905 Einstein proposed that light itself is quantized. A beam of light of frequency $f$ consists of particles, **photons**, each carrying energy

$$\boxed{E = hf = \frac{hc}{\lambda},} \tag{6}$$

with $h$ the same constant Planck had found from blackbody radiation. In the photoelectric effect a single electron absorbs a single photon, taking all of its energy at once. The electron uses at least $\phi$ to escape the metal, and the most kinetic energy it can leave with is

$$\boxed{K_{\max} = hf - \phi.} \tag{7}$$

Electrons that lose some energy on their way out of the metal leave with less.

Each observation now follows directly.

* **Intensity.** A more intense beam has more photons per second, not more energetic ones. It ejects more electrons, giving a larger current, but $K_{\max}$ depends only on $f$.

* **Cutoff frequency.** If $hf < \phi$, no single photon has enough energy to free an electron, however many photons arrive. The cutoff is at $hf_c = \phi$, or equivalently at the cutoff wavelength $\lambda_c = hc/\phi$.

* **No delay.** The energy arrives concentrated in one photon rather than spread over the wavefront, so an electron can be ejected as soon as the first photon arrives.

* **Linear dependence on $f$.** Combining $K_{\max} = eV_s$ with $K_{\max} = hf - \phi$,

$$V_s = \frac{h}{e} f - \frac{\phi}{e}. \tag{8}$$

  This is a straight line with slope $h/e$, the same for every metal, and intercept $f_c = \phi/h$, which depends on the metal. This is exactly Figure 2(b).

Einstein's prediction was tested carefully by Robert Millikan, who expected to prove it wrong. His 1916 measurements of $V_s$ versus $f$ confirmed the straight line, and its slope gave a value of $h$ in agreement with Planck's. The same constant appears in two completely different experiments. Einstein received the 1921 Nobel Prize for this work, and Millikan the 1923 prize.

### Example: photoelectrons from potassium

*Ultraviolet light of wavelength* 300 *nm shines on potassium ($\phi = 2.2$ eV). Find the maximum kinetic energy of the ejected electrons, the stopping voltage, and the longest wavelength that ejects electrons.*

The photon energy is

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{300 \text{ nm}} = 4.13 \text{ eV}, \tag{9}$$

so

$$K_{\max} = hf - \phi = 4.13 \text{ eV} - 2.2 \text{ eV} = 1.9 \text{ eV}, \tag{10}$$

and the stopping voltage is $V_s = K_{\max}/e = 1.9$ V. The cutoff occurs when the photon energy just equals the work function:

$$\lambda_c = \frac{hc}{\phi} = \frac{1240 \text{ eV} \cdot \text{nm}}{2.2 \text{ eV}} = 560 \text{ nm}. \tag{11}$$

Light with wavelength longer than 560 nm (yellow, orange, red) ejects no electrons from potassium, however bright it is.

**Example: photons from a laser pointer**

*A red laser pointer emits* 1.0 *mW at* 650 *nm. How many photons does it emit per second?*

Each photon carries

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{650 \text{ nm}} = 1.91 \text{ eV} = (1.91)(1.60 \times 10^{-19} \text{ J}) = 3.06 \times 10^{-19} \text{ J}. \tag{12}$$

The number per second is the power divided by the energy per photon:

$$\frac{1.0 \times 10^{-3} \text{ J/s}}{3.06 \times 10^{-19} \text{ J}} = 3.3 \times 10^{15} \text{ photons per second}. \tag{13}$$

Individual photons carry so little energy that an ordinary beam contains an enormous number of them, which is why light normally appears continuous.